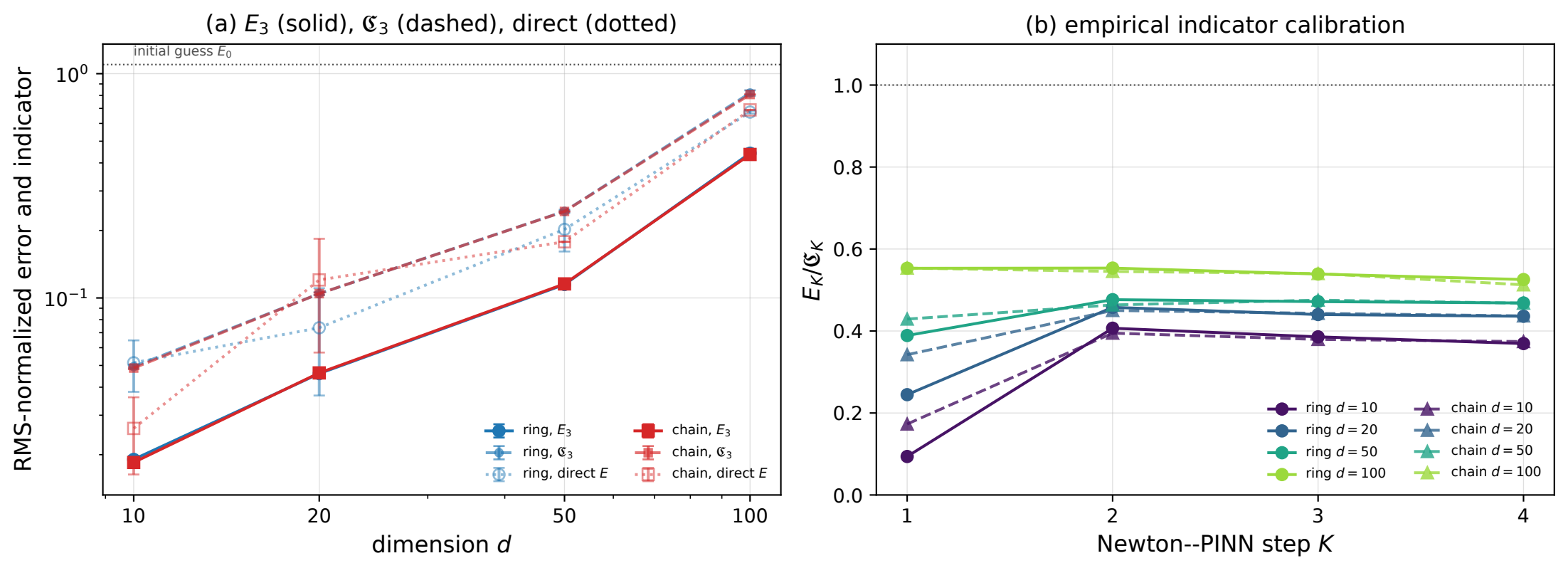
(a) E_3 (solid), $\mathfrak{C}_3$ (dashed), direct (dotted)
initial guess E_0
RMS-normalized error and indicator
10^0
10^{-1}
10
20
50
100
dimension d
ring, E_3
ring, $\mathfrak{C}_3$
ring, direct E
chain, E_3
chain, $\mathfrak{C}_3$
chain, direct E
(b) empirical indicator calibration
$E_K/\mathfrak{C}_K$
1.0
0.8
0.6
0.4
0.2
0.0
1
2
3
4
Newton--PINN step K
ring $d = 10$
ring $d = 20$
ring $d = 50$
ring $d = 100$
chain $d = 10$
chain $d = 20$
chain $d = 50$
chain $d = 100$